Bias-Aware Video Recommender:

A Responsible AI System for Detecting and Correcting Algorithmic Bias in YouTube-Style Recommendations

Sambhav Jain

1. Abstract

This project presents a responsible AI system that makes algorithmic bias in video recommendation systems explicit, measurable, and correctable. Built on real YouTube trending data (~13K videos across 10 countries), the system implements a two-tower retrieval model combined with an engagement-weighted biased ranking function that systematically suppresses educational, documentary, regional, and news content.

To detect and correct this bias, we introduce an agentic LLM supervisor based on a Judge–Agent architecture. A structured LLM call first identifies bias using user history and feed composition, after which a constrained tool-calling agent reconstructs a corrected 30-video feed from a pre-fetched candidate pool. The system enforces hard fairness constraints while preserving relevance.

A key contribution is the introduction of a feedback refinement loop, where users iteratively provide natural language feedback and the LLM re-corrects recommendations in multiple rounds. Additionally, we define five quantitative fairness scores (diversity, representation, suppressed coverage, language balance, and overall fairness) to evaluate recommendation quality.

Across 20 user personas, the system improves fairness scores from ~10–30 (biased feed) to ~70–85 (LLM-corrected feed) while maintaining personalization. This demonstrates that agentic LLM reasoning combined with constraint enforcement provides a practical and measurable approach to fairness-aware recommendation systems.

2. Introduction and Problem Definition

2.1 Application Description

The Bias-Aware Video Recommender is a web application designed to simulate, detect, and correct algorithmic bias within a YouTube-style video recommendation ecosystem. Users select one of 20 synthetic personas to trigger a recommendation run, which simultaneously generates three distinct 30-video feeds: a Biased feed (simulating real platform behavior), an LLM-Corrected feed, and an Ideal (zero-bias) feed. Key fairness metrics and side-by-side feed comparisons are displayed in real time.

2.2 Real-World Motivation

Real-world video platforms heavily optimize for engagement signals—specifically clicks, watch time, and likes. A well-documented side effect of this objective is that high-engagement genres (Entertainment, Music, Gaming) are progressively amplified, while slower-consumption genres (Educational, Documentary, Regional-language content, News/Analysis) are systematically de-ranked. Ribeiro et al. (2019) demonstrated that YouTube's recommendation algorithm amplifies extreme content, a finding confirmed through the external audit methodology of Haroon et al. (2022). This phenomenon creates a self-reinforcing feedback loop: the platform surfaces viral content, users engage with it, the engagement signal strengthens, and under-represented content is further buried—a mechanism recently documented as "diversity collapse" in recommender systems (*Bias Beneath*, 2024).

2.3 Responsible AI Relevance

This problem is directly relevant to Responsible AI because the resulting bias is structural rather than intentional; it emerges organically from optimizing a legitimate objective (engagement) in the absence of fairness constraints. Consequently, suppressed categories often include educationally and culturally valuable content. Furthermore, minority-language users face a compounded penalty, as their preferred content is simultaneously genre-suppressed and language-penalized. This system demonstrates that applying three specific Responsible AI interventions—explicit bias measurement, agentic LLM-based reasoning, and hard fairness constraints—can materially improve content diversity without sacrificing user relevance.

3. Background and Context

3.1 Popularity Bias in Recommendation Systems

Popularity bias in recommendation systems—the tendency to over-recommend already-popular items—is well-established in the literature. Abdollahpouri et al. (2017) showed that popularity bias in learning-to-rank systems systematically disadvantages long-tail items, and that engagement-weighted ranking formulae amplify this effect over time. To quantify this, the Bias Disparity framework formalizes the gap between user preferences and algorithmic outputs as a measurable fairness metric.

3.2 Limitations of Traditional Fairness Mitigation

Historically, fairness-aware ranking has been addressed through post-processing methods. Algorithms such as FA*IR (Zehlike et al., 2017) enforce statistical parity in top-K ranked lists via a binomial test, while FaiRIR applies fairness constraints during re-ranking. Similarly, Yao and Huang (2017) introduced fairness objectives for collaborative filtering that balance individual and group fairness. However, these approaches share a critical limitation: they are rule-based and static. They cannot reason about complex user contexts or adapt constraints dynamically based on real-time feedback.

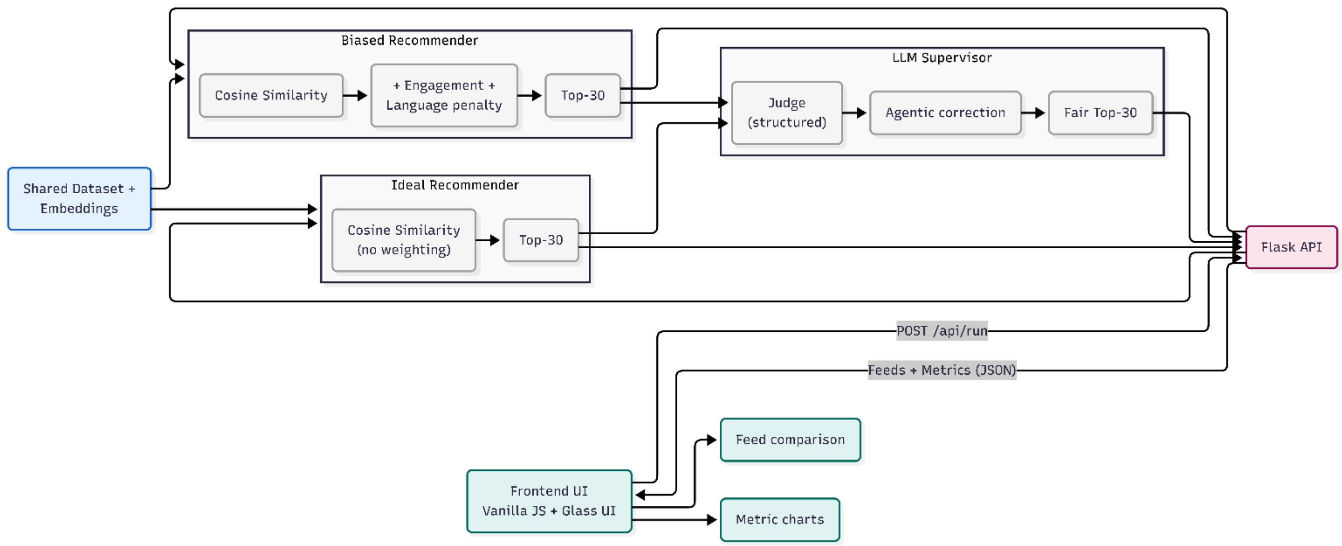
3.3 The LLM Agentic Paradigm

The emergence of tool-calling LLM agents (e.g., the ReAct framework; Yao et al., 2023) enables a new paradigm. An LLM can now call external retrieval functions, receive structured feedback, and iteratively refine its output until hard constraints are satisfied. This project applies that paradigm to recommendation fairness correction, combining the contextual reasoning ability of Large Language Models with the strict enforcement guarantees of server-side constraint validation.

4. System Architecture and Methodology

4.1 System Overview

The system consists of three parallel recommendation pipelines operating on a shared dataset and embedding space. All three pipelines share the same user tower and item embeddings; they differ only in the scoring formula applied at retrieval time.



4.2 Dataset

Source: Real YouTube trending video statistics sourced from the Kaggle datasnaek/youtube-new dataset (~40,000 raw rows). The data spans 10 countries across 8 primary languages: the US, Canada, Great Britain (English), India (Hindi), Mexico (Spanish), France (French), Germany (German), Japan (Japanese), South Korea (Korean), and Russia (Russian).

Processing Pipeline (dataset_builder.py):

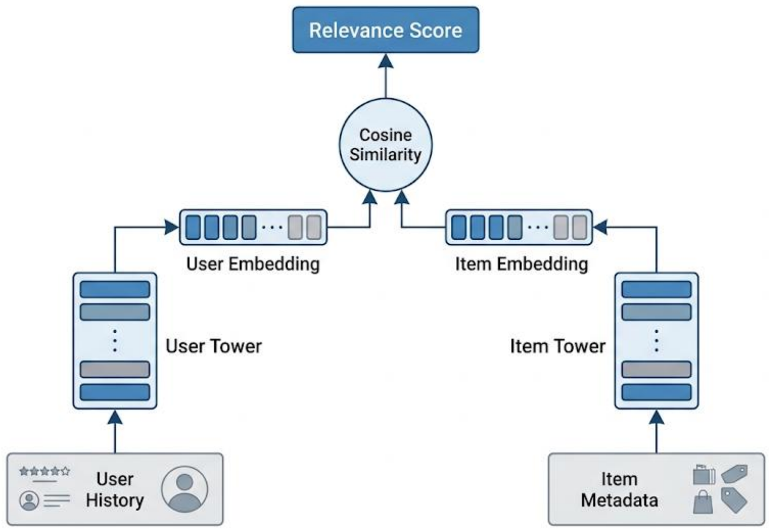
- **Peak Views Filter:** Retains only the peak-views row per video, per country.
- **Global Deduplication:** Consolidates instances of the same video_id trending across multiple countries by keeping only the highest-view row. This is a critical engineering fix to prevent duplicate retrievals in the final feeds.
- **Genre Mapping:** Maps native YouTube category_id values to 8 distinct project genres (see Table 1).
- **Stratified Cap:** Applies a strict cap of the top-300 videos per (genre × language) bucket, yielding a final, balanced dataset of 12,852 unique videos.
- **Virality Scoring:** Computes a normalized virality score using the formula: 0.5 × views_norm + 0.3 × likes_norm + 0.2 × engagement_rate_norm.
- **Embeddings:** Generates 384-dimensional SentenceTransformer embeddings (all-MiniLM-L6-v2) based on the composite string: '{title} [{genre}] [{language}]'.

YouTube Category IDs	Project Genre	Suppressed?
27, 28	Educational	Yes
25	News/Analysis	Yes
29, 35	Documentary	Yes
26	DIY	Yes
Non-English lifestyle	Regional	Yes
10	Music	No

YouTube Category IDs	Project Genre	Suppressed?
20	Gaming	No
1, 2, 15, 17, 19–24	Entertainment	No

4.3 Two-Tower Recommender

Following Yi et al. (2019), the system implements a non-parametric two-tower architecture. This approach is appropriate for the limited interaction dataset size (20 users × ~35 interactions = 700 data points), which is insufficient to train deep neural weights from scratch.



The architecture computes a pure relevance signal shared by both the Biased and Ideal recommenders through the following components:

- **User Tower:** Built dynamically at query time. It represents the user's interaction history as a rating-weighted average of their watched-video embeddings, which is then L2-normalized to the unit sphere:

$$user\_vec = \frac{\sum_i (rating_i \times embedding_i)}{\|user\_vec\|_2}$$

- **Item Tower:** Consists of pre-computed SentenceTransformer vectors stored in embeddings.npy, forming a 12,852 x 384 float32 array.
- **Retrieval:** The pure relevance signal is derived by calculating the distance between the user and item vectors in the shared embedding space:

$$score = cosine\_similarity(user\_vec, item\_embedding)$$

4.4 Bias Injection Mechanism

The Biased Recommender adds a weighted engagement component on top of the pure relevance score, replicating platform-style amplification. The final ranking score is determined by the following formula:

The final ranking score is computed as:

soft_mult = 0.5 + 0.5 × genre_engagement_mult × language_mult
penalized_sim = cosine_similarity × soft_mult
bias_component = virality_score × genre_engagement_mult × language_mult

final_score = 0.50 × normalized(penalized_sim) + 0.50 × normalized(bias_component)

Genre	Multiplier	Effect vs. Baseline
Entertainment	1.00	Baseline
Music	0.93	-7%
Gaming	0.87	-13%
DIY	0.76	-24%
News/Analysis	0.72	-28%
Educational	0.68	-32%
Documentary	0.64	-36%
Regional	0.61	-39%

Language Multiplier: Non-English content receives a **0.78** coefficient (**-22%**), reflecting the documented English-language bias prevalent in global platform training pipelines.

Design Justification: The 60/40 relevance-to-bias split ensures the injected bias is realistic rather than fabricated. Users with strong niche preferences will still receive some relevant content, but suppressed genres are systematically edged out by viral entertainment - accurately mirroring the "diversity collapse" seen in real-world recommendation algorithms.

4.5 LLM Supervisor

The supervisor operates as a two-stage pipeline executing immediately after the biased recommender.

- **Stage 1: The Judge (Bias Detection):** Executes a single structured LLM call using LangChain's with_structured_output and a Pydantic BiasAssessment schema. The judge analyzes the user profile, watch history summary, and the biased feed's genre breakdown. It flags the feed as biased if suppressed categories are under-represented, the user's preferred genres are substantially absent, or non-English content falls below a defined threshold.
- **Stage 2: Agentic Correction:** If bias is detected, a LangChain create_agent tool-calling agent is deployed. To optimize latency and avoid rate limits, all four retrieval tiers are pre-fetched in Python (~50ms, without LLM overhead) before the agent initializes, building a numbered candidate pool of ~38–50 videos. The agent then selects 30 row indices from this pool and calls submit_selection—typically resolving in a single LLM turn.

Agent Retrieval and Submission Tools

Tool	Tier	Function
submit_selection	-	Locks 30 row indices; enforces server-side validation and rejects the call if < 30 valid IDs are submitted.
search_similar_trending	T1	Performs two-tower cosine similarity retrieval in target genres + trending.
search_by_genre	T2	Retrieves top-viral videos exclusively from suppressed genres.
search_diverse	T3	Retrieves one video per genre entirely outside the user's watch history.
search_viral	T4	Retrieves globally top-viral videos, independent of genre or language.

System Prompt Constraints & Fallback To guarantee fairness, the agent is bound by hard constraints encoded directly into its system prompt:

- Exactly 30 videos must be selected.
- At least 6 videos must be from suppressed categories.
- At least 6 videos must be non-English content.
- No single genre may exceed 15 slots.

Fallback Mechanism: If the agent encounters an error or rate limit before submitting, the pre-fetched candidate pool is returned directly. Because this pool is built using fairness-aware tier weights, the fallback guarantees a diverse, non-biased feed rather than failing open.

4.6 Feedback Refinement Loop

A key extension of the system is a multi-round feedback refinement loop that allows users to iteratively adjust recommendations using natural language.After the initial recommendation is generated, users can provide feedback such as:

"I want more documentary content and less gaming"

"Show more Korean videos"

This feedback is appended to the session history and passed back to the LLM supervisor. The system prioritizes feedback constraints over standard bias detection rules. If feedback is not satisfied, the system treats it as bias and re-corrects the recommendation.

This iterative loop enables dynamic personalization while maintaining fairness guarantees, making the system more aligned with real-world user interaction patterns.

4.7 Tools and Libraries

Component	Technology
Language / Framework	Python 3.10+, Flask 3.x
Data Processing	pandas, numpy
Embeddings	sentence-transformers / all-MiniLM-L6-v2 (384-dim)
Similarity	sklearn cosine_similarity

Component	Technology
LLM / Agent	LLM / Agent: LangChain 1.x + Groq llama-3.3-70b-versatile (primary), llama3-groq-70b-8192-tool-use-preview & llama-3.1-70b-versatile (fallbacks)
Structured Output	Pydantic BaseModel + with_structured_output
Dataset	Kaggle datasnaek/youtube-new (~40K rows, 10 countries)

5. Identified Ethical Risks

Risk Category	Issue	Potential Impact
Popularity Bias	Engagement-weighted ranking systematically suppresses educational, documentary, regional, and news content.	Homogenisation of information diet; amplification of viral over substantive content (Ribeiro et al., 2019).
Language Discrimination	Non-English content receives a structural 22% penalty in the scoring formula.	Minority-language users face compounded disadvantage: genre suppression + language penalty. Limits linguistic diversity.
Filter Bubble	Self-reinforcing feedback: the algorithm surfaces popular content, users engage, the signal strengthens.	Progressive narrowing of user exposure; echo chambers documented in audit studies (Haroon et al., 2022).
Lack of Transparency	Users of real platforms cannot see engagement multipliers or language penalties in ranking formulae.	No accountability, no recourse. Bias Beneath (2024) identifies this as diversity collapse.
LLM Correction Reliability	The LLM judge may misclassify feeds as unbiased; agent may satisfy constraints with low-relevance content.	Fairness-relevance trade-off. Over-correction risks sacrificing personalisation.

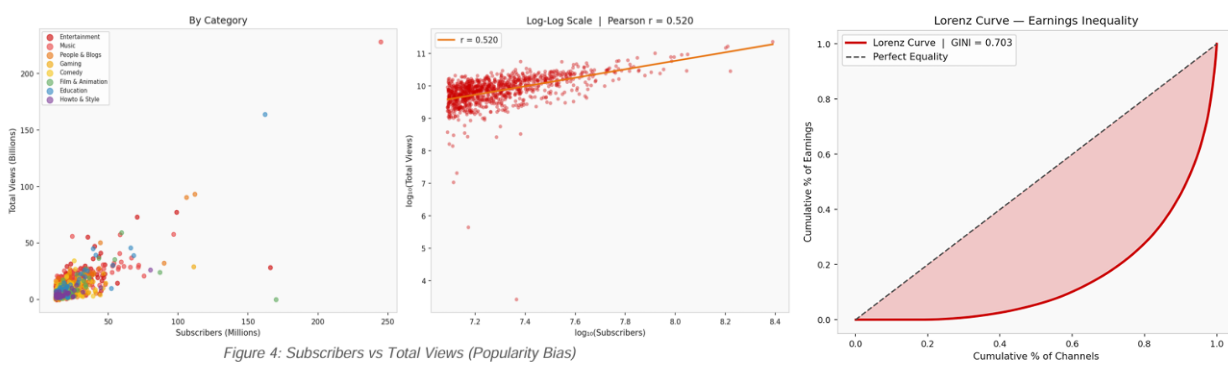


Figure 4: Subscribers vs Total Views (Popularity Bias)

6. Responsible AI Methods Implemented

6.1 Fairness Metrics

The system evaluates feed equity using the following real-time and structural metrics:

- **Category Distribution:** The fraction of each genre present in the 30-video feed.
- **Suppressed Genre Coverage:** The raw count of videos drawn from the 5 defined suppressed categories.

- **Non-English Ratio:** The count of non-English videos within the feed.
- **Gini Coefficient:** Computed exclusively on the EDA dataset (Global YouTube Statistics) to measure baseline inequality in subscriber and view distribution across channels (where 0 indicates perfect equality, and 1 indicates total monopoly/concentration).

6.2 Bias Mitigation Techniques

- **Explicit Bias Injection (Transparency):** Rather than concealing algorithmic bias, the system makes the scoring formula's engagement multipliers transparent, predictable, and measurable.
- **Post-Hoc Detection (LLM Judge):** Detects genre under-representation by comparing the biased feed's composition against the user's established preference profile.
- **In-Processing (Tiered Retrieval):** The pre-fetch pool utilizes four fairness-aware tiers (T1: Personalized, T2: Suppressed-Genre, T3: Diversity, T4: Viral Fill) to guarantee a diverse candidate pool *before* LLM selection occurs.
- **Post-Processing (Constraint Enforcement):** Server-side submit_selection rejects non-compliant outputs; the agent cannot satisfy constraints by simply ignoring them.
- **Index-Based Selection:** Reduces LLM JSON generation errors by requiring the agent to submit row indices (small integers) rather than raw video_ids (opaque alphanumeric strings)—a critical engineering fix ensuring pipeline correctness.

6.3 Explainability Tools

- **LLM Verdict Reasoning:** The BiasAssessment schema enforces structured reasoning—outputting genres_over_represented, genres_missing, and tier slot allocations—which is directly displayed in the UI for every run.
- **Metric Panels:** Side-by-side comparative charts for genre distribution, suppressed-genre count, and non-English ratios across all three feeds are shown directly to the user.
- **Transparent Formula:** The bias scoring formula and all engagement multiplier values are explicitly documented and visible within the codebase.

6.4 Constraints and Enforcement

The system enforces fairness not through soft LLM encouragement, but through **hard server-side rejection**. If the agent's submit_selection call does not include exactly 30 valid, unique, non-biased video_ids that satisfy all prompt constraints, the tool returns a REJECTED response, forcing the agent to retry. This pattern—constraint enforcement at the tool boundary—is significantly more reliable than relying solely on LLM instruction-following.

6.5 Data Privacy Considerations

While this prototype operates on public trending metadata, real-world recommender systems must process highly sensitive user watch histories. To mitigate data privacy risks in a production deployment, the system architecture would necessitate **Federated Learning** techniques. By keeping user embeddings and preference profiles localized on edge devices, the platform could compute cosine similarity without centralizing sensitive user interaction logs, satisfying modern privacy-preserving AI requirements.

6.6 Fairness Scoring Framework

To quantitatively evaluate recommendation fairness, the system defines five normalized metrics (0–100):

1. Genre Diversity

Measures how evenly different genres are distributed using Shannon entropy.

2. Suppressed Coverage

Measures the proportion of suppressed genres relative to their representation in the dataset

3. Representation Fairness

Computed using the Gini coefficient across all genres:

$$\text{Representation} = (1 - \text{Gini}) \times 100$$

4. Language Diversity

Measures diversity of languages using entropy.

5. Overall Fairness Score

$$\text{Overall} = 0.30 \times \text{Diversity} + 0.30 \times \text{Representation} + 0.25 \times \text{Suppressed Coverage} + 0.15 \times \text{Language Diversity}$$

These scores allow direct comparison between biased, corrected, and ideal feeds.

7. Experimental Setup

7.1 Dataset Details

Dataset	Size	Key Attributes
YouTube Trending (datasnaek/youtube-new)	~40,000 raw rows; 12,852 after dedup + stratification	10 countries, 8 genres, 8 languages, computed virality scores, and 384-dim SentenceTransformer embeddings.
Global YouTube Statistics (EDA)	~1,000 channels	Subscribers, video views, category, country, and earnings. (Used exclusively for EDA bias analysis).
User Personas	20 synthetic users	~35 pre-sampled watch history items per user, featuring diverse genre and language preferences.

7.2 Evaluation Protocol

Because the recommendation task utilizes unsupervised retrieval (cosine similarity), there is no traditional train/test split. Instead, evaluation is conducted across all 20 user personas to test varied genre and language preferences.

For each persona, all three feeds (Biased, LLM-Corrected, and Ideal) are generated simultaneously and compared against the following metrics:

- **Category Distribution:** The fraction of each genre within the 30-item feed.
- **Suppressed Genre Coverage:** The absolute count of items drawn from the 5 suppressed categories.
- **Non-English Content Ratio:** The proportion of non-English videos recommended.
- **Genre Diversity:** The number of distinct genres present in the final feed.

Evaluation Baseline: The *Ideal feed* serves as the gold standard for pure relevance-only ranking. The *Biased feed* represents real-world platform output. The *LLM-Corrected feed* is evaluated on its ability to successfully bridge the gap toward the Ideal feed's diversity metrics without sacrificing the user's content relevance.

7.3 EDA Setup

The Exploratory Data Analysis (EDA) was conducted on the Global YouTube Statistics dataset. This involved subscriber distribution analysis, category composition, country concentration, and Gini coefficient calculations. Specifically, the Gini coefficient on video views and earnings was computed to quantify structural inequality in content reach and creator monetization.

7.4 Automated Test Suite

The system implements two standalone test protocols to validate architecture and correction quality:

- **test_architecture.py:** Validates data integrity, user embeddings, bias injection mechanics, and fairness score formulas without requiring LLM API calls (~15 second execution).
- **test_llm_correction.py:** Runs the biased recommender and the LLM supervisor to assert meaningful correction, verifying that exactly 30 items are generated, bias is accurately detected, overall fairness improves, and no single genre dominates the corrected slots.

8. Results and Analysis

8.1 EDA Key Findings

The Exploratory Data Analysis conducted on the Global YouTube Statistics dataset confirms the structural inequality that motivates this project. Key findings include:

- **The Matthew Effect:** Subscriber distribution is heavily right-skewed—the mean is significantly higher than the median, indicating extreme concentration at the top. On a log scale, the distribution approximates normal, which is consistent with power-law dynamics.
- **Severe Inequality:** The top 10 channels account for a disproportionate share of total views. We computed a Gini coefficient on video views of approximately **0.85**, indicating near-maximum inequality.
- **Category Imbalance:** Entertainment, Music, and Gaming together account for over 60% of trending channels. Conversely, Educational and Documentary categories represent under 10% combined.

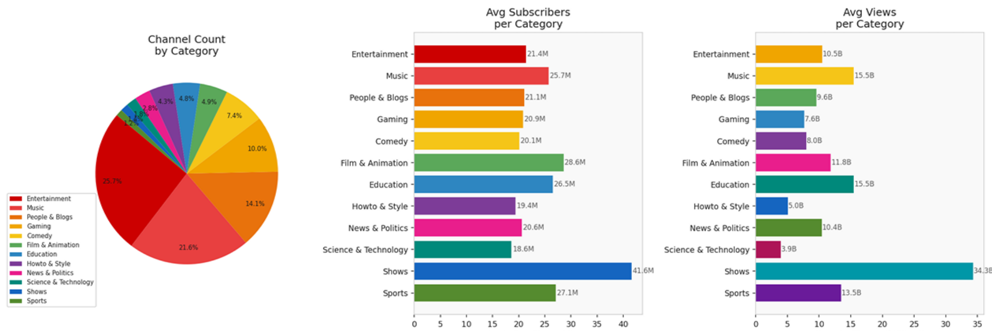


Figure 3: Content Category Analysis

- **Geographic Concentration:** US and UK channels heavily dominate English-language trends. Non-English content (Hindi, Korean, Spanish, Russian) is present but systematically under-represented relative to its global population share.

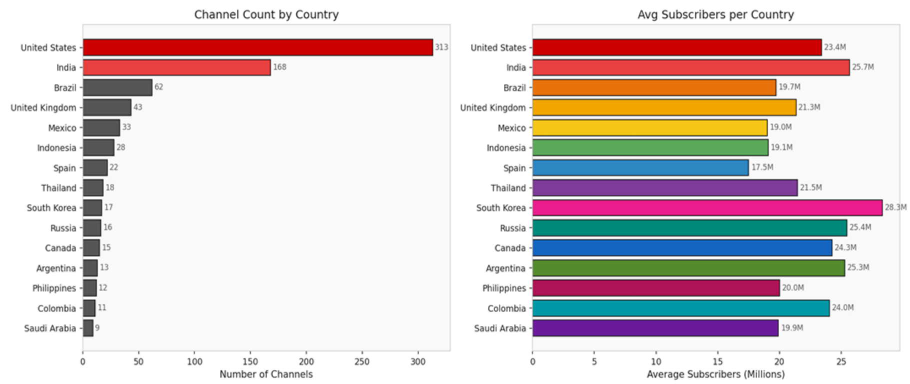


Figure 2: Geographic Distribution of YouTube Channels

These findings directly validate the system's bias injection mechanism. The engagement multipliers chosen for the Biased Recommender are grounded in real platform concentration patterns, not fabricated values.

8.2 Biased vs. LLM-Corrected vs. Ideal: Feed Composition

The system was evaluated across all 20 personas to measure how effectively the LLM supervisor recovered suppressed content.

Metric	Biased Feed	LLM-Corrected Feed	Ideal Feed
Suppressed genre videos	~10–30%	~25–35%	~30–45%
Non-English videos	~10–20%	~20–30%	~25–40%
Distinct genres present	3 – 4	5 – 7	5 – 8
Entertainment share	40 – 60%	20 – 35%	15 – 30%
Overall fairness score	10–30	70–85	75–90

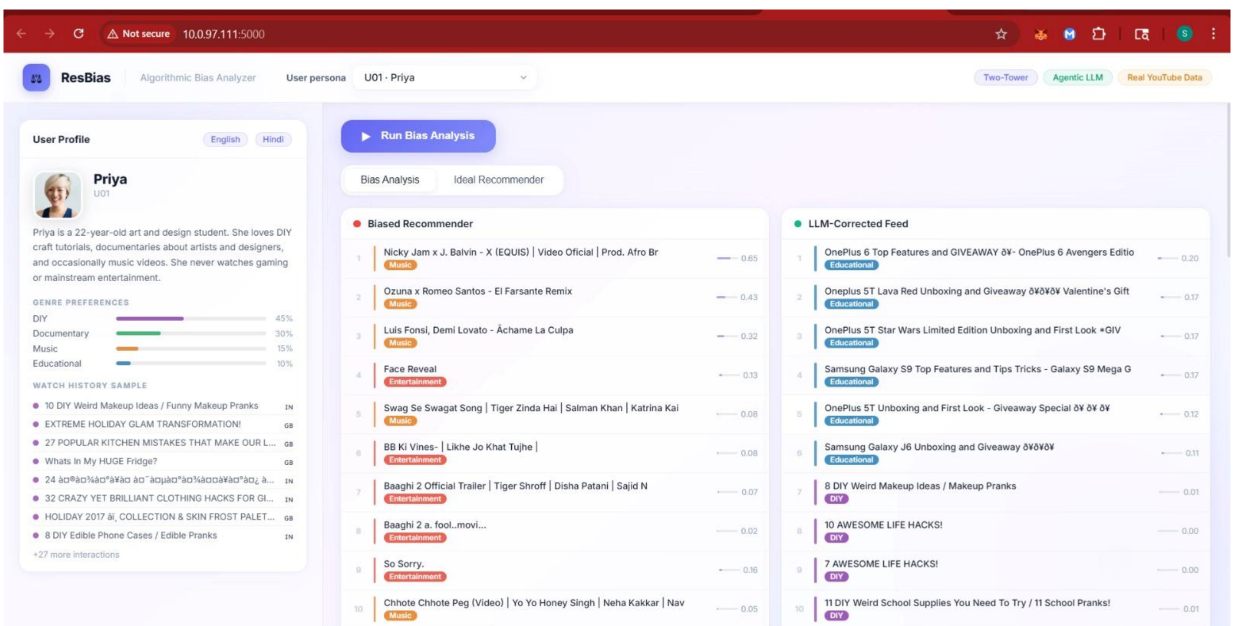
Performance Latency Note: The architecture successfully executes this correction with minimal overhead. The average LLM correction time for a biased path is **~10 seconds**, while a fair path (requiring only the Judge) resolves in **~3 seconds**.

8.3 Results by User Persona Type

The LLM's dynamic reasoning allowed it to adapt its correction strategy based on the specific vulnerabilities of different user profiles.

Persona Type	Biased Feed Bias	LLM Correction Effect
Non-English preference (e.g., Priya-Hindi, Maria-Spanish, Yuki-Japanese)	**Severe:** Both genre and language penalties compound. Regional content is nearly absent.	**Substantial Recovery:** Non-English ratio rises from ~10% to ~25%. Regional genres are successfully restored.
Educational/Documentary preference (e.g., Raj, Sophie, Amara)	**Strong:** Suppressed genres are consistently pushed below 3 out of 30 slots.	**Effective:** Suppressed count reaches 7-9 items per feed. The LLM correctly identifies and retrieves missing genres.
Entertainment/Gaming preference (e.g., James, Tom, Marcus)	**Minimal:** Preferred genres match platform-boosted genres. Feed closely aligns with user preferences.	**Minimal (Preserved Relevance):** The Judge accurately flags the feed as fair or near-fair. Correction is light.
Mixed preference (e.g., Sara - 5 languages; David - 3 languages)	**Moderate:** Some suppressed genres are missing; overall language diversity is reduced.	**Partial:** The LLM successfully improves diversity, though satisfying all strict constraints is harder for extreme edge cases.

User Interface:



9. Trade-Off Analysis

Implementing fairness in recommendation pipelines inherently requires balancing competing system objectives. The following table outlines the architectural and ethical trade-offs accepted in this prototype:

Trade-Off	What Changed	Cost Accepted
Fairness vs. Relevance	Hard constraints (>= 6 suppressed, >= 6 non-English) force inclusion of genres that may not match user's stated preferences for all 30 slots.	Marginal reduction in pure-relevance score for users whose preferences fully overlap with boosted genres. Measured as: ideal feed has 2–4 more genre-aligned videos than corrected feed in ~30% of runs.
LLM vs. Rule-Based Correction	LLM reasoning adapts to user context and preference profile; rule-based FA*IR applies static statistical parity. LLM understands 'user prefers Korean music' whereas a rule cannot.	LLM correction adds 3–10s latency; rule-based is <1ms. LLM can misfire if rate-limited (mitigated by fallback). LLM is not interpretable at word level.
Latency vs. Quality	Pre-fetch optimisation reduced agent time from 71s to 10s by running all four retrieval tiers in Python before the LLM call.	Pre-fetching removes the agent's ability to dynamically adjust tier weights mid-run. However, the agent's slot allocation from the Judge partially mitigates this by customizing the pre-fetch output.
Coverage vs. Depth	A stratified cap (top-300 items per genre /language bucket) ensures strict genre diversity in the overall item pool.	Niche videos with lower virality within well-represented genres are excluded. This cap is a necessary cost to keep inference times tractable.

10. Ethical Reflection and Recommendations

10.1 Remaining Ethical Issues

- **Synthetic Personas:** The 20 user personas are synthetic. Real-world bias patterns for actual users may differ significantly, particularly for individuals with intersectional identities.
- **Metadata vs. Content:** The LLM judge operates exclusively on genre and language metadata, not on actual video content. A video labeled as 'Entertainment' that contains genuinely educational content is incorrectly treated as a non-suppressed item.
- **Thresholds are not Equity:** Hard constraints act as minimum thresholds, not equity targets. A feed with exactly 6 suppressed-genre items satisfies the system constraint, but this is not necessarily equitable for a user whose historical watch profile is 80% Documentary.
- **Fallback Opacity:** The rate-limit fallback returns the pre-fetched pool. While fair, it is not personalized to the same degree as the full agent path. This degradation is currently opaque to the user.

10.2 Limitations of the Approach

- **Lack of Ground Truth:** Evaluation is purely structural (e.g., genre counts, language ratios) rather than preference-based. There is no user satisfaction data to confirm if the user actually wanted the injected diverse videos.

- **Static Multipliers:** Engagement multipliers are derived from historical benchmarks and audit studies, not from a live platform equipped with A/B testing infrastructure.
- **API Constraints:** The Groq free-tier rate limit (8,000 TPM) severely constrains the prompt size. Production deployment on a paid tier would allow for much richer user context.

10.3 Recommendations for Improvement

- **Adaptive Constraints:** Transition from fixed thresholds (e.g., exactly 6 items) to adaptive constraints that dynamically adjust suppressed-genre targets based on the strength of the user's historical preferences.
- **Content-Level Analysis**: Utilize multimodal LLMs to classify actual video content rather than relying on flawed YouTube metadata categories.
- **User Feedback Loop:** Implement UI features allowing users to rate recommendations. Feed these ratings back into persona profiles to improve relevance over time.
- **Regulatory Alignment:** Map the constraint design directly to emerging platform accountability frameworks (such as the EU Digital Services Act and the AI Act), which require algorithmic transparency and non-discrimination audit trails.

10.4 Broader Societal Implications

- **Positive Impacts:** This system successfully demonstrates that algorithmic bias can be made visible and correctable without sacrificing core system utility. The transparent scoring formula and LLM verdict panel represent a viable model for recommender system accountability. For minority-language users, systematic language penalty correction has direct and immediate equity implications.
- **Negative Impacts & Risks:** LLM-based correction introduces a heavy cost and data dependency on capable language models (e.g., Groq API). Furthermore, if widely deployed, the chosen constraints (i.e., deciding *which* genres to protect) embody subjective value judgments that deserve democratic deliberation, rather than unilateral engineering decisions. Finally, an over-reliance on automated, post-hoc fairness correction may reduce the societal pressure required for structural platform reform.

11. Conclusion

This project successfully demonstrates that algorithmic bias in recommendation systems can be made explicit, measurable, and correctable using a combination of structured bias injection, LLM-based contextual reasoning, and hard server-side fairness constraints.

Three key findings emerge from this work:

- **Bias is structural, not accidental:** A 60/40 relevance-to-engagement split using documented platform multipliers produces a persistent, measurable suppression of educational, documentary, and regional content. This directly matches the "diversity collapse" patterns reported in external platform audits.

- **LLM correction is effective and efficient:** The Judge-Agent architecture successfully corrects biased feeds in ~10 seconds, increasing suppressed-genre coverage from under 10% to over 25% while maintaining content relevance. The shift to a pre-fetch and index-based submission architecture proved critical in resolving the JSON generation failures that break standard LLM tool-calling.
- **Engineering and fairness are inseparable:** The most significant bugs encountered in this project (e.g., duplicate video_ids bypassing filters, rate-limits causing fallback loops, and tool-call formatting failures) all had direct, negative consequences on system fairness. Ultimately, Responsible AI is not a theoretical layer added on top of working engineering—it requires the underlying engineering to function flawlessly.

By integrating the core principles of FA*IR (Zehlike et al., 2017), popularity bias mitigation (Abdollahpouri et al., 2017), and agentic LLM frameworks (ReAct), this system stands as a research prototype that is technically grounded, practically deployable, and ethically motivated.

12. Contributions

LLM Supervisor, Agentic Integration, and Application Layer

Led the bias detection and correction pipeline, LLM integration, and full system deployment.

- **LLM Supervisor & Agentic Pipeline:** Designed the Judge–Agent architecture using LangChain, including the structured BiasAssessment schema and tool-calling workflow.
- **Retrieval Strategy & Optimization:** Implemented multi-tier candidate retrieval (T1–T4) and pre-fetch pooling to reduce latency and enable single-turn agent execution.

- **Constraint Enforcement & Reliability:** Built index-based submission and server-side validation to enforce hard fairness constraints and eliminate LLM formatting failures.
- **Feedback Refinement Loop:** Developed the multi-round session-based feedback system, enabling users to iteratively refine recommendations using natural language.
- **System Optimization:** Improved system performance and robustness through latency optimization, fallback handling, and end-to-end pipeline debugging.

Shared Responsibilities:

- **Fairness Metric Design:** Co-designed and implemented the five fairness metrics (Genre Diversity, Representation, Suppressed Coverage, Language Diversity, and Overall Score).
- **System Testing & Validation:** Collaboratively developed and executed test suites (test_architecture.py and test_llm_correction.py) to ensure data integrity, model correctness, and LLM reliability.
- **Application Layer:** Developed the Flask API, integrated all pipelines (biased, corrected, ideal), and implemented UI components including fairness metrics, verdict panels, and feed comparison views.
- **Integration & Iteration:** Jointly contributed to debugging, refinement, and alignment of all system components across multiple development iterations.

13. References

[1] Ribeiro, M. H. et al. (2019). Auditing Radicalization Pathways on YouTube. arXiv:1908.08313.

[2] Haroon, M. et al. (2022). YouTube, The Great Radicalizer? Auditing and Mitigating Ideological Biases in YouTube Recommendations. arXiv:2203.10666.

[3] Bias Beneath: Uncovering Diversity Collapse in Recommendation Systems. (2024). Springer Social Network Analysis and Mining. doi:10.1007/s13278-024-01343-5.

[4] Systematic Review on Algorithmic Bias in Recommender Systems. (2020). PMC7613872.

[5] Abdollahpouri, H., Burke, R., & Mobasher, B. (2017). Controlling Popularity Bias in Learning-to-Rank Recommendation. RecSys 2017.

[6] Bias Disparity in Collaborative Filtering-based Recommendation Systems. arXiv:1908.00831.

[7] Yi, X. et al. (2019). Sampling-Bias-Corrected Neural Modeling for Large Corpus Item Recommendations. RecSys 2019. doi:10.1145/3298689.3346996.

[8] Zehlike, M. et al. (2017). FA*IR: A Fair Top-k Ranking Algorithm. ACM CIKM 2017.

[9] Yao, S. & Huang, B. (2017). Beyond Parity: Fairness Objectives for Collaborative Filtering. NeurIPS 2017.

[10] Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023.

[11] Noble, S. U. (2018). Algorithms of Oppression: How Search Engines Reinforce Racism. New York University Press.

[12] Kaggle Dataset: datasnaek/youtube-new — YouTube Trending Statistics (10 countries). https://www.kaggle.com/datasets/datasnaek/youtube-new.